Ho Ko

AI/ML SOFTWARE ENGINEER

CONTACT

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SKILLS

Languages

Python · C/C++ · Swift · Java · Kotlin · JavaScript · x86 Assembly · MATLAB

ML Lifecycle

Model Training · Distillation · Quantization & Compression · Inference Engines · On-Device Deployment · Scalable Serving

On-Device & Web

WebGPU · Three.js · Transformers.js · MediaPipe · WebRTC · Local LLMs (Gemma, LFM)

Domains

NLP · Computer Vision · Healthcare · Astronomy · Weather Forecasting · Economics

EDUCATION

M.S. Computer Science
University of Southern California — AI & ML specialization
2021–2023 · GPA 4.0/4.0

M.S. Electrical & Computer Eng.
University of Waterloo — Machine Learning & AI
2020–2021 · GPA 3.9/4.0

B.S. Computer Science
National Central University, Taiwan
2016–2020 · GPA 3.8/4.0

AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

EXPERIENCE

Machine Learning Engineer · Google — 2026–Present

- Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
- **Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.

Software Engineer · Google — 2023–2026

- Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
- Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI and Google AI Core, reaching 1B+ users**.
- Shipped the **Google AI Edge Gallery** app on Android and iOS.

Research Collaborator · Google DeepMind — 2024–2025

- Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
- Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.

Research Collaborator · Google Health & Fitbit — 2024–2025

- Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.

Software Engineer · MyCareLinq — 2023

- Built **NLP pipelines and a home-health-care knowledge graph** from unstructured health documentation.
- Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.

Software Engineer Intern · Google — 2022

- Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
- Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.

Research Assistant · Caltech (SURF) — 2018

- Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

SELECTED PUBLICATIONS

- *Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection* — ICLR 2025, Google DeepMind.
- *Opportunistic Hypertension Detection from Wearable PPG* — medRxiv 2025, Google Health & Fitbit.
- *A Type IIb Supernova (ZTF18aalrxas)* — ApJL 878:L5, 2019.

SELECTED PROJECTS

- **AI Mesh** — P2P network of on-device AI agents over WebRTC; each browser tab runs a local LLM + MCP knowledge base and gossips queries across the mesh.
- **The Match Maker** — Webcam multiplayer game running Gemma 3n fully client-side via WebGPU at ~15 tok/s with MediaPipe face mesh.
- **Tetris AI in x86 Assembly** — Self-learning heuristic AI in ~2,000 lines of hand-written x86, averaging 50,000 cleared lines.